

Shape-based embeddings of time series

How to learn time series embeddings that account for their shape ?

Shape analysis for time series

Germain, T., Gruffaz, S., Truong, C., Durmus, A., & Oudre, L.

Advances in neural information processing systems (NeurIPS), 2024.

March 13th, 2026

L2S seminar

Motivation

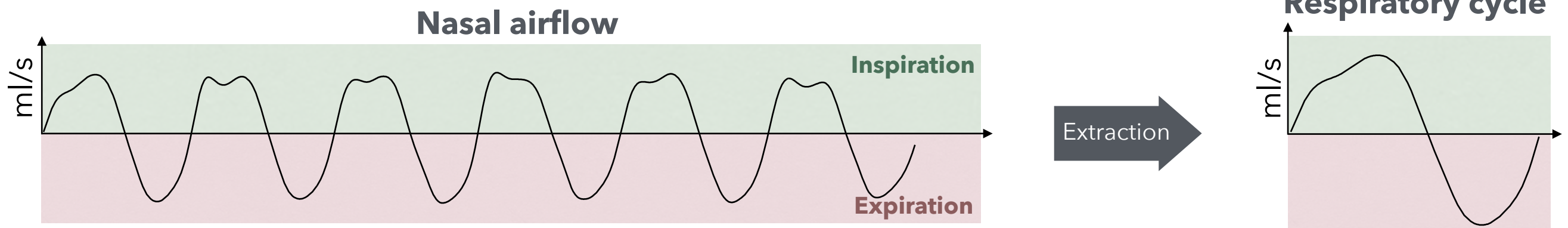
Inspired by a use case: mouse ventilation analysis

Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes.

Experimental protocol: For each genotype (WT, ColQ):



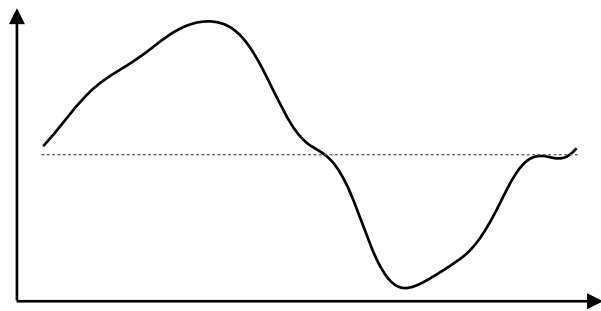
First insight: Time series with elementary patterns:



Inspired by a use case: mouse ventilation analysis

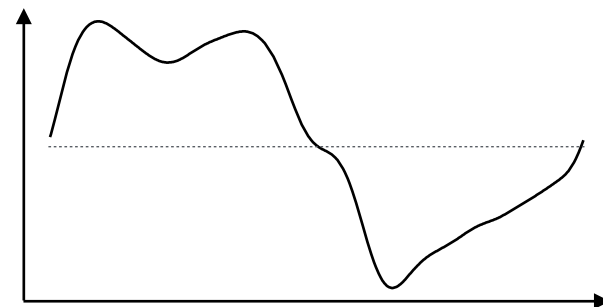
Biologist's insights: Shapes of respiratory cycles accounts for physiological states.

WT (before)



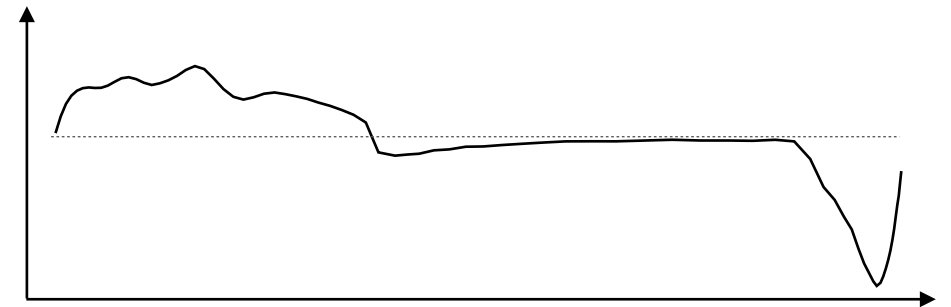
« Healthy mouse's respiratory cycles »

ColQ (before)



« The two bumps during inspiration indicate fine-motor control difficulties »

WT (after)



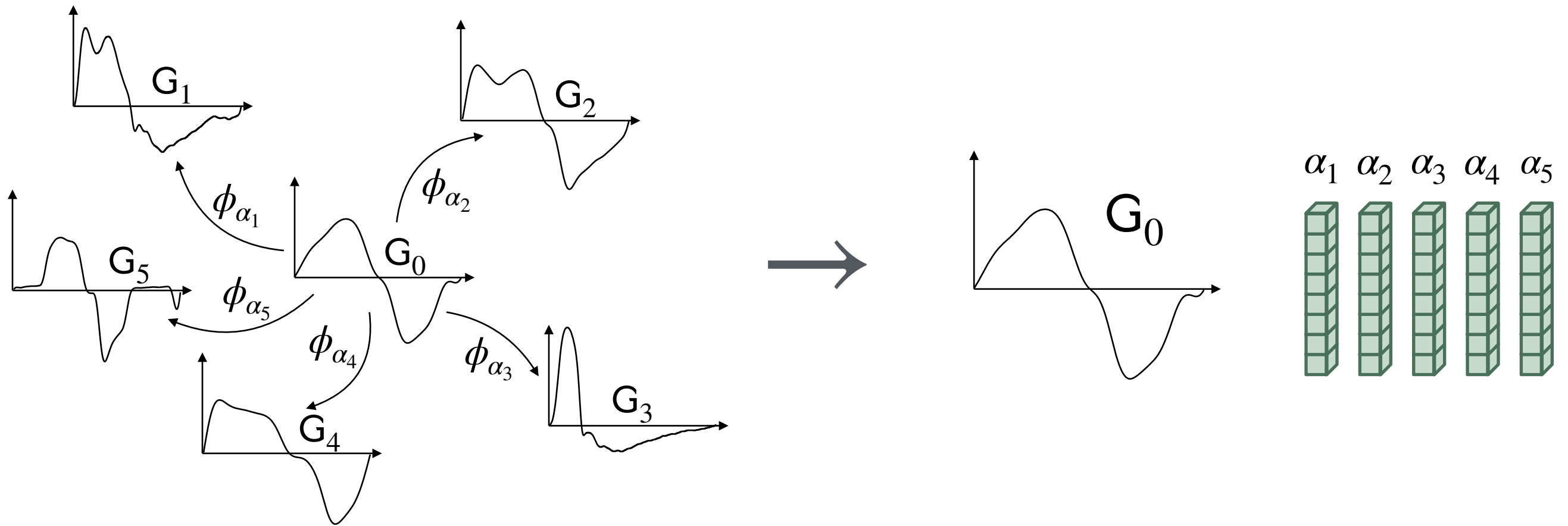
- « The constrained flow during inspiration indicates airways obstructions »
- « The long pause indicates tetany, its duration is important »

Transcription to modeling & requirements:

- Embed time series as **deformations of a referent time series**
- **Two-way mapping** between temporal and embedding spaces
- **Compatible with statistical analysis** (identifiable model on a vector space)
- Work for **small datasets** with **irregularly sampled** and **variable-length** time series

Embedding framework outlines

Embedding outlines



Input: $\left(s_j : \mathbf{I}_j \mapsto \mathbb{R}^d\right)_{j \in [N]}$

1. The shape of s_j is $\mathbf{G}_j = \{(t, s_j(t)) : t \in \mathbf{I}_j\}$.
2. $\mathbf{G}_j$ is represented as the deformation of a reference graph $\mathbf{G}_0 = \{(t, s_0(t)) : t \in \mathbf{I}\}$, i.e.:

$$\mathbf{G}_j \approx \phi_{\alpha_j} \cdot \mathbf{G}_0 = \{\phi_{\alpha_j}(t, s_0(t)) : t \in \mathbf{I}\},$$
 where $\phi_{\alpha_j} : \mathbb{R}^{d+1} \mapsto \mathbb{R}^{d+1}$ is a diffeomorphism parametrized by $\alpha_j \in \mathbb{R}^m$.

Output: a graph of reference $\mathbf{G}_0 \subset \mathbb{R}^{d+1}$, deformation parameters $(\alpha_j)_{j \in [N]} \in (\mathbb{R}^m)^N$

How to learn G_0 and α_j 's ?

An optimization problem: $G_0 \subset \mathbb{R}^{d+1}$ and $(\alpha_j)_{j \in [N]} \in (\mathbb{R}^m)^N$ are learned by solving the empirical Fréchet mean problem:

$$\arg \min_{G_0, (\alpha_j)_{j \in [N]}} \frac{1}{N} \sum_{j \in [N]} \left(\underbrace{d_G^2(\phi_{\alpha_j} \cdot G_0, G_j)}_{\text{data attachment}} + \underbrace{\lambda d_\Phi^2(\text{Id}, \phi_{\alpha_j})}_{\text{regularization}} \right)$$

Open questions:

1. How to parametrized the diffeomorphisms $(\phi_{\alpha_j})_{j \in [N]}$?
2. What distances (d_G, d_Φ) should be considered ?

Remark: Description of d_G is not covered, intuitively it can be think as a maximum mean discrepancy (MMD) between graphs seen as measures [Kal17].

Generating diffeomorphisms: The LDDMM framework [Beg05]

How to parametrized diffeomorphisms ?

Theorem: Global diffeomorphism for small perturbation to the identity.

Let $f : \mathbb{R}^n \mapsto \mathbb{R}^n$ be a continuously differentiable (C^1) function such that:

$$\|Df(x)\|_\infty < 1, \quad \text{for any } x \in \mathbb{R}^n.$$

Then the map:

$$F(x) \triangleq x + f(x)$$

is a C^1 -diffeomorphism from $\mathbb{R}^n$ to $\mathbb{R}^n$.

Intuition: Diffeomorphism build by composition:

$$F = F_N \circ F_{N-1} \circ \dots \circ F_1$$

with $F_i = Id + f_i$ and $\|Df_i\|_\infty < 1$, for any $i \in [N]$.

Our case: Diffeomorphism defined by infinite composition, i.e. it is the flow of an ordinary differential equation (ODE).

Building diffeomorphisms with LDDMM [Beg05]

Theorem [Gla05]: Assuming $v \in L^2([0,1], V)$ a time-varying velocity field in $\mathbb{R}^n$, where V is an RKHS with some regularity assumptions [Gla05]. For any $x_0 \in \mathbb{R}^n$, the differential system:

$$\frac{dX(\tau)}{d\tau} = v_\tau(X(\tau)) \quad \text{with} \quad X(0) = x_0,$$

has a unique solution defined for all $\tau \in [0,1]$.

At any time $\tau \in [0,1]$, **the flow application:**

$$\phi_v^\tau : x_0 \in \mathbb{R}^n \mapsto x_0 + \int_0^\tau v_r(\phi_v^r(x_0)) dr \in \mathbb{R}^n$$

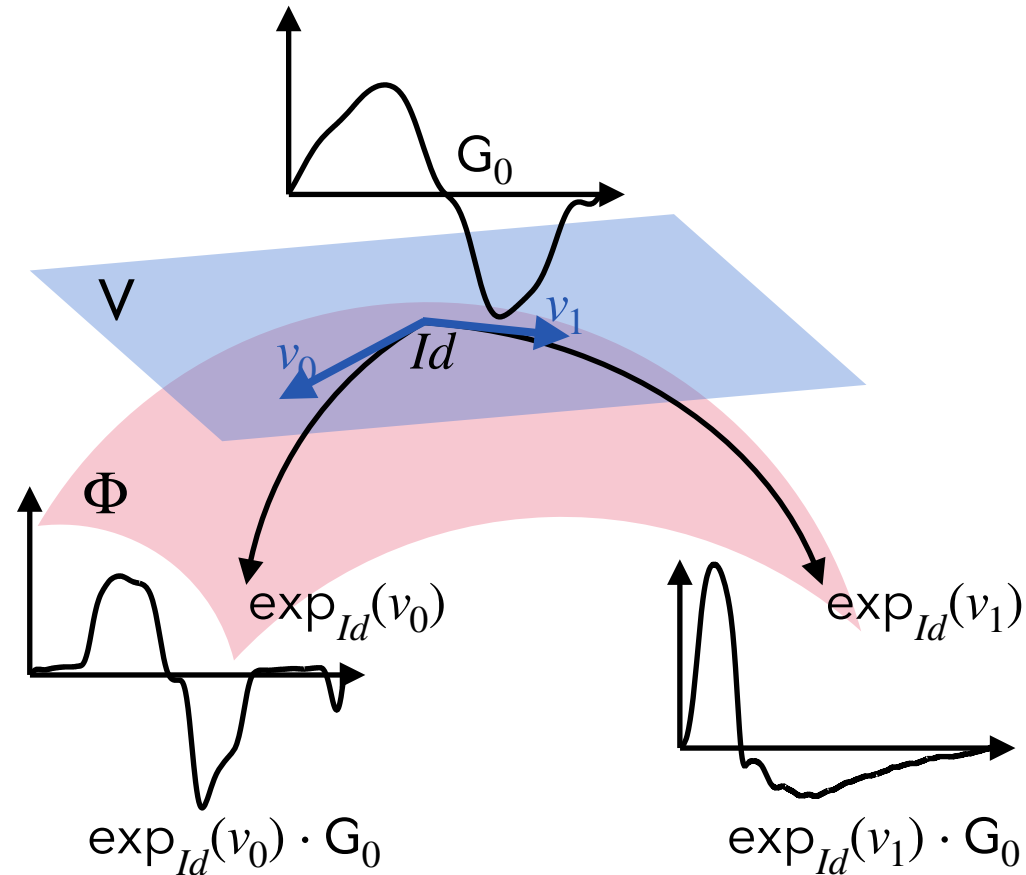
is a **diffeomorphism**.

A metric group: The group of diffeomorphisms $\Phi \triangleq \{\phi_v^1 \mid v \in L^2([0,1], V)\}$ is metrizable such that for any $\phi \in \Phi$:

$$d_\Phi^2(Id, \phi) = \inf_{v \in L^2([0,1], V)} \left\{ \int_0^1 \|v_\tau\|_V^2 d\tau \mid \phi_v^1 = \phi \right\},$$

the infimum is reached with a v^* and it conserves its norm along its geodesic path i.e.: $\|v_\tau^*\|_V = \|v_0^*\|_V, \forall \tau \in [0,1]$.

Geodesic shooting



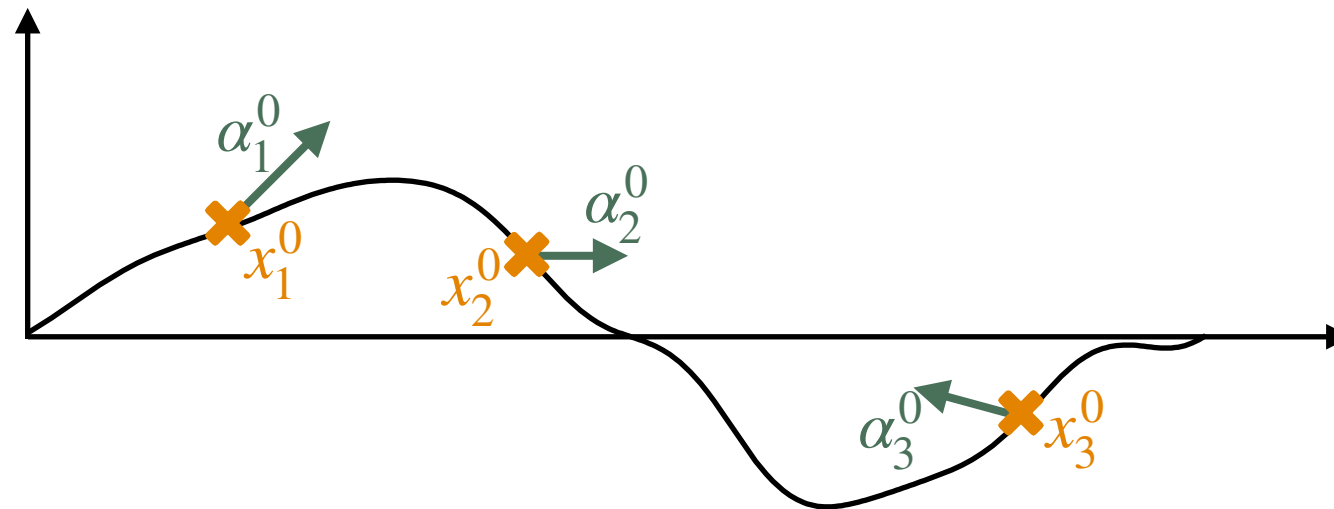
An exponential map: By denoting $\tau \mapsto \rho_{v_0}(\tau)$ the geodesic starting from Id with initial conditions $v_0 \in V$, the exponential map is [Mil06]:

$$\exp_{Id} : v_0 \in V \mapsto \rho_{v_0}(1) \in \Phi \quad \text{and} \quad d_{\Phi}^2(Id, \exp_{Id}(v_0)) = \|v_0\|_V^2$$

Important remarks:

- The exponential map is locally bijective.
- The geodesic path is uniquely define by the initial conditions.

Geodesic shooting in practice



Let denote $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^{n^2}$ the kernel of the RKHS $\mathcal{V}$.

Initial velocity field: Given N_0 control points $X_0 = (x_k^0)_{k \in [N_0]} \in (\mathbb{R}^n)^{N_0}$, and momentums $\alpha = (\alpha_k^0)_{k \in [N_0]} \in (\mathbb{R}^n)^{N_0}$, the initial velocity field is,

$$v_0 : x \in \mathbb{R}^n \mapsto \sum_{k \in [N_0]} K(x_k^0, x) \alpha_k^0 \in \mathbb{R}^n .$$

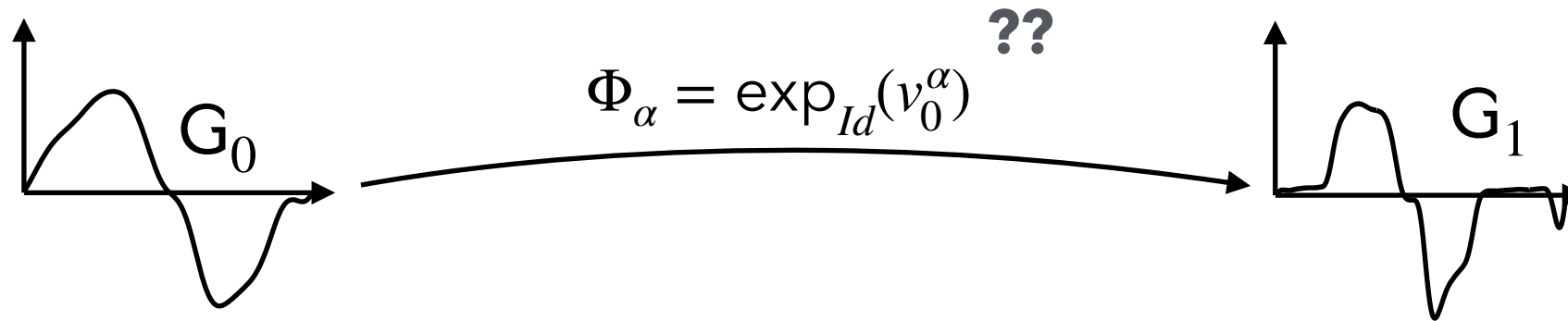
Along the geodesic path: For any $\tau \in [0,1]$, the velocity field verifies:

$$v_\tau : x \in \mathbb{R}^n \mapsto \sum_{k \in [N_0]} K(x_k(\tau), x) \alpha_k(\tau) \in \mathbb{R}^n ,$$

Geodesic equations:

$$\begin{cases} \frac{dx_k(\tau)}{d\tau} = v_\tau(x_k(\tau)) \\ \frac{d\alpha_k(\tau)}{d\tau} = - \sum_{l \in N_0} d_{x_k(\tau)} K(x_k(\tau), x_l(\tau)) \alpha_l(\tau)^\top \alpha_k(\tau) \end{cases} \quad \text{with} \quad \begin{cases} x_k(0) = x_k^0 \\ \alpha_k(0) = \alpha_k^0 \end{cases} \quad \forall k \in [N_0] .$$

Shape registration



Question: How to learn the initial momentum $\alpha \in (\mathbb{R}^n)^{N_0}$ encoding the diffeomorphism mapping G_0 to G_1 ?

Minimize: $\text{loss}(\alpha) \triangleq d_G^2(\exp_{Id}(v_0^\alpha) \cdot G_0, G_1) + \lambda \|v_0^\alpha\|_V^2$

Protocol: Given G_0 and G_1 do:

Set $\alpha = 0$

While did not converge:

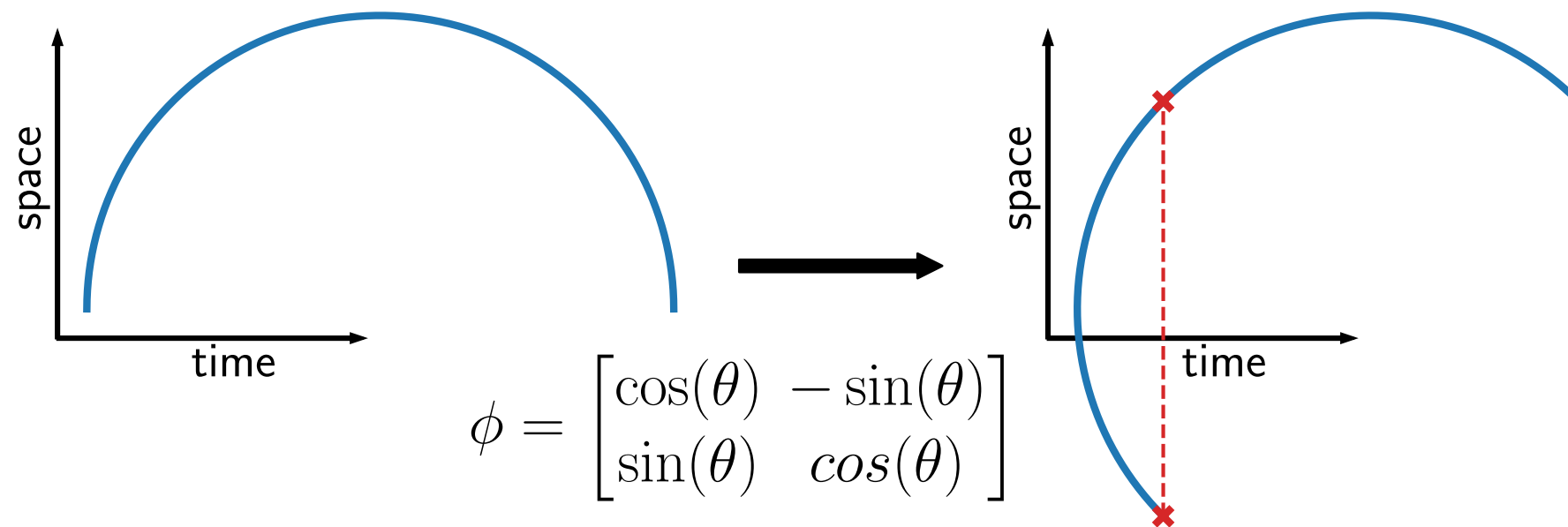
1. Perform geodesic shooting: $\exp_{Id}(v_0^\alpha)$ (ODE resolution: RK4)
2. Compute $\text{loss}(\alpha)$
3. Backward gradient (classical autodiff or reverse ODE with the adjoint method)

Return α

Application to time series

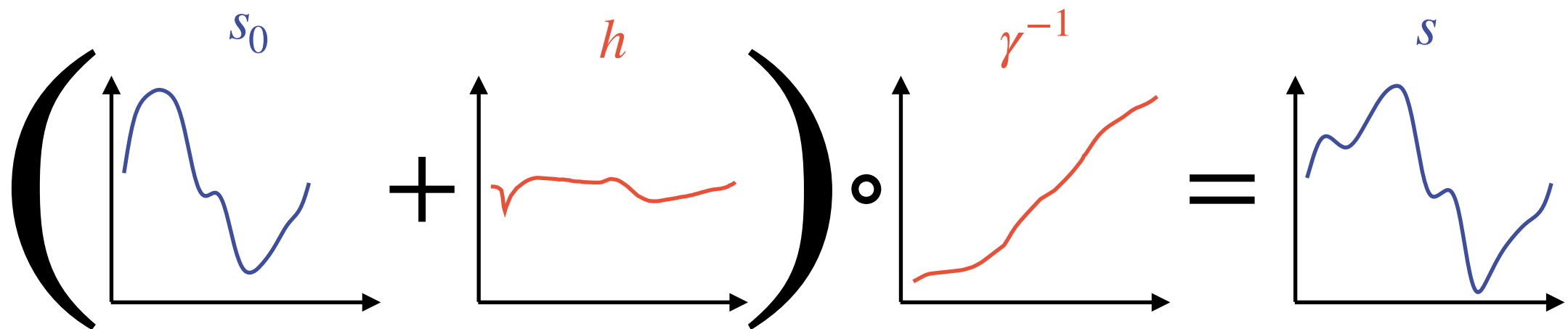
Respecting the graph structure of time series

Problem: A general diffeomorphism does not necessarily preserve the structure of a time series graph.



Deformation of times series

Intuition: Let $s_0 : \mathbf{I} \mapsto \mathbb{R}^d$ and $s : \mathbf{J} \mapsto \mathbb{R}^d$, the diffeomorphic deformation ϕ mapping s_0 to s should be seen as distortion $h : \mathbf{I} \mapsto \mathbb{R}$ and a time parametrization $\gamma^{-1} : \mathbf{J} \mapsto \mathbf{I}$ such that: $\phi \cdot \mathbf{G}(s_0) = \mathbf{G}((s_0 + h) \circ \gamma^{-1}) = \mathbf{G}(s)$.

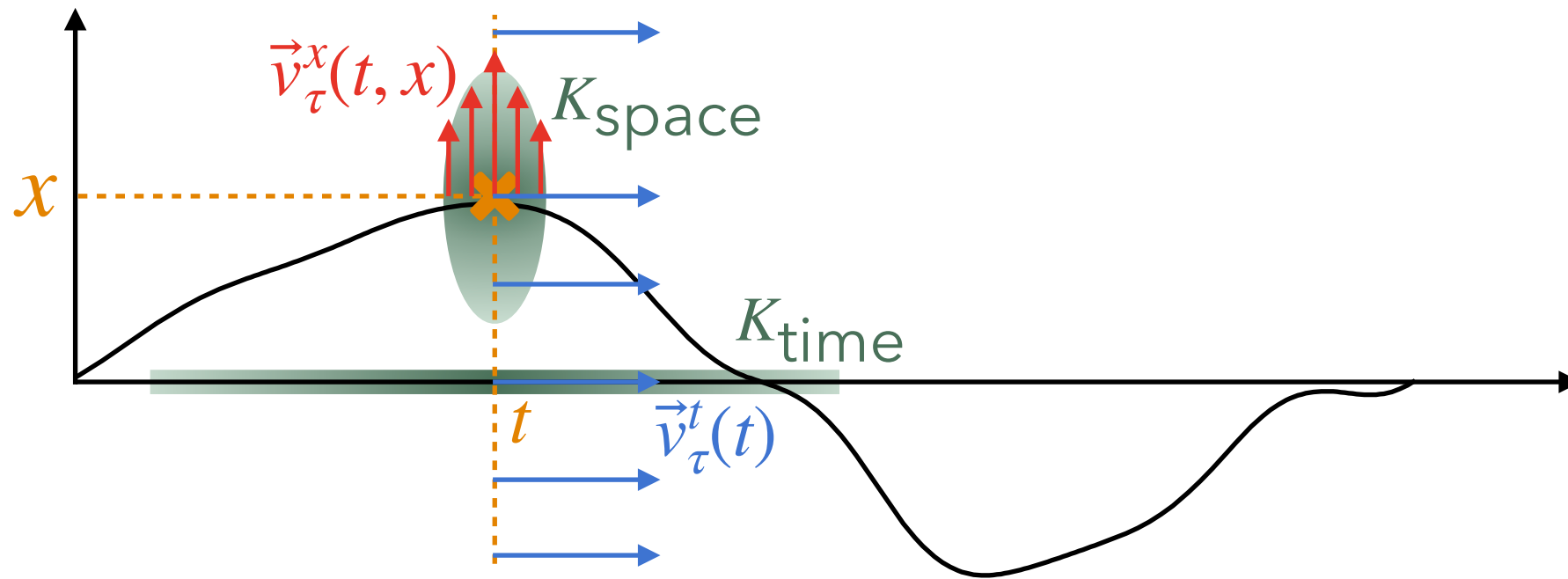


Theorem (representativity): Any continuously differentiable time series $s_0 : \mathbf{I} \mapsto \mathbb{R}^d$ and $s : \mathbf{J} \mapsto \mathbb{R}^d$, there exists deformations $\Psi_\gamma : (t, x) \in \mathbb{R}^{d+1} \mapsto (\gamma(t), x) \in \mathbb{R}^{d+1}$ with $\gamma \in \mathbf{D}(\mathbb{R})$, and $\Pi_f : (t, x) \in \mathbb{R}^{d+1} \mapsto (t, f(t, x)) \in \mathbb{R}^{d+1}$ with $f \in \mathbf{C}^1(\mathbb{R}^{d+1}, \mathbb{R}^d)$, such that:

$$\phi_{\gamma, f} \cdot \mathbf{G}(s_0) = \mathbf{G}(s) \quad \text{with} \quad \phi_{\gamma, f} = \Psi_\gamma \circ \Pi_f,$$

Moreover, for any $\bar{\gamma} \in \mathbf{D}(\mathbb{R})$, and $\bar{f} \in \mathbf{C}^1(\mathbb{R}^{d+1}, \mathbb{R}^d)$, $\phi_{\bar{\gamma}, \bar{f}} \cdot \mathbf{G}(s_0)$ is the graph of a continuously differentiable time series.

A kernel for time series



The proposed kernel: Let K_G be a kernel defined for any $(t, x), (t', x') \in (\mathbb{R}^{d+1})^2$,

$$K_G((t, x), (t', x')) = \begin{pmatrix} c_0 K_{\text{time}} & 0 \\ 0 & c_1 K_{\text{space}} \end{pmatrix} \quad \text{with} \quad \begin{cases} K_{\text{time}} = K_{\sigma_{T,0}}^{(1)}(t, t') \\ K_{\text{space}} = K_{\sigma_{T,1}}^{(1)}(t, t') K_{\sigma_x}^{(d)}(x, x') I_{\mathbb{R}^d} \end{cases}$$

parametrized by $\sigma_{T,0}, \sigma_{T,1}, \sigma_x > 0$ and the constants $c_0, c_1 > 0$.

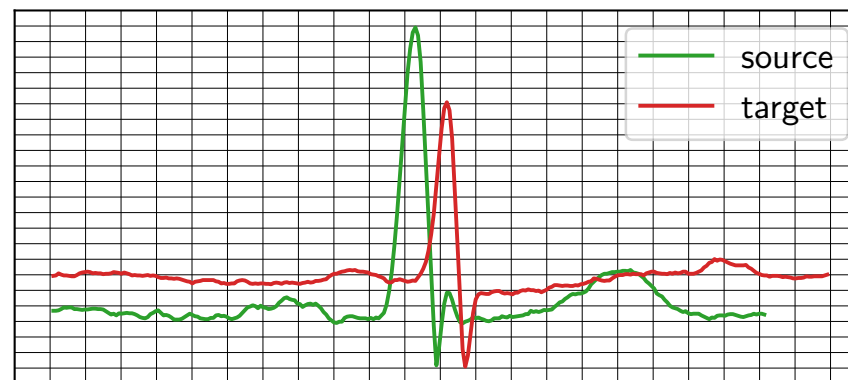
Lemma: For any initial velocity field $v_0 \in V$, the RKHS associated to K_G , and for any $\tau \in [0,1]$, there exist a time parametrization γ_τ and a distortion h_τ such that:

$$\exp_{Id}(\tau v_0) \cdot G(s_0) = G((s_0 + h_\tau) \circ \gamma_\tau^{-1}).$$

Experiments

Example of transport

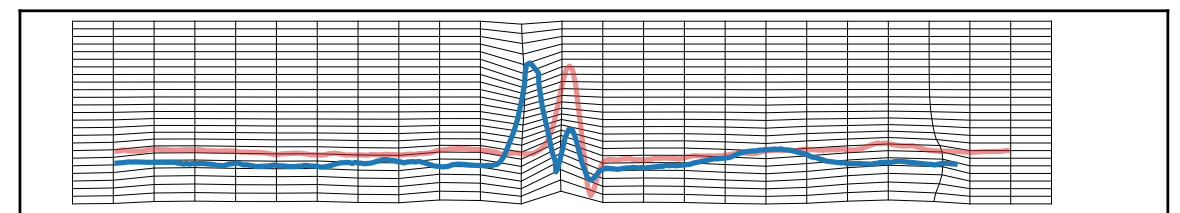
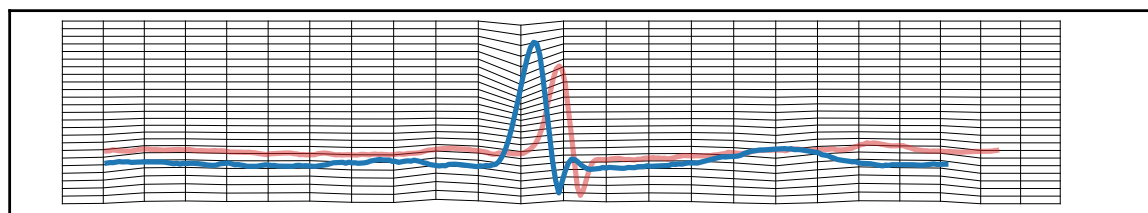
ECG heartbeat



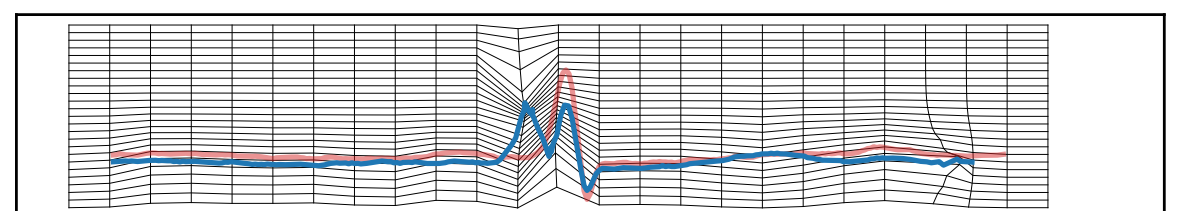
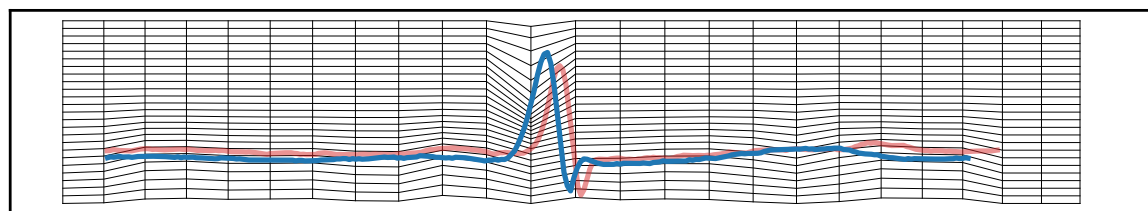
TS-LDDMM (ours)

LDDMM

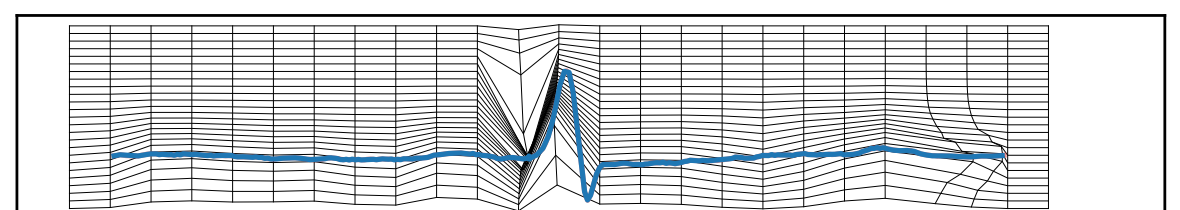
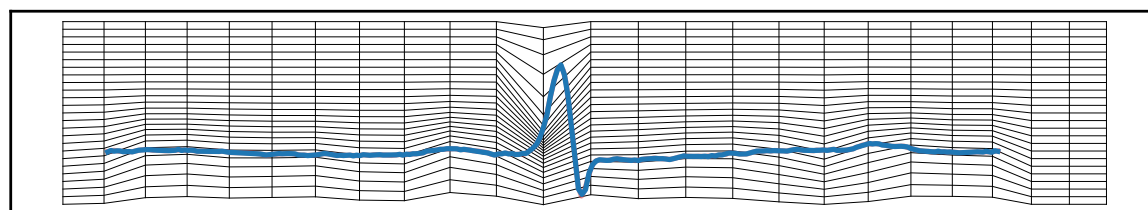
$\tau = 0.3$



$\tau = 0.6$



$\tau = 1.0$



Robustness to irregular sampling

Experimental protocol:

- Task: Classification with different scenarios of irregular sampling.
- Datasets: 15 UCR/UEA datasets.
- Metric: f1-score.
- Algorithms: TS-LDDMM, LDDMM and 8 deep learning algorithms.

Results:

First & second best performers. TS-LDDMM is the best performer on three out of four dropping rate.

Methods	Regular		30 % dropped		50 % dropped		70 % dropped	
	F1-score	Rank	F1-score	Rank	F1-score	Rank	F1-score	Rank
RNN (1999)	0.64 ± 0.21	6.2	0.53 ± 0.23	6.6	0.48 ± 0.21	7.2	0.44 ± 0.21	6.07
LSTM (1997)	0.61 ± 0.29	6.0	0.57 ± 0.29	6.27	0.53 ± 0.25	6.07	0.51 ± 0.29	5.27
GRU (2014)	0.71 ± 0.26	4.2	0.68 ± 0.28	4.27	0.66 ± 0.28	3.73	<u>0.59 ± 0.28</u>	<u>3.67</u>
MTAN (2021)	0.59 ± 0.28	7.13	0.58 ± 0.28	5.8	0.54 ± 0.29	5.33	0.51 ± 0.28	5.0
MIAM (2022)	0.48 ± 0.35	6.93	0.42 ± 0.33	8.27	0.47 ± 0.31	6.93	0.35 ± 0.31	7.6
ODE-LSTM (2020)	0.63 ± 0.24	6.0	0.57 ± 0.25	6.53	0.51 ± 0.24	7.27	0.45 ± 0.23	6.73
Neural SDE (2019)	0.48 ± 0.28	7.67	0.47 ± 0.26	7.47	0.45 ± 0.27	7.13	0.45 ± 0.25	6.0
Neural LNSDE (2024)	0.7 ± 0.27	<u>3.87</u>	0.68 ± 0.29	<u>4.0</u>	<u>0.67 ± 0.25</u>	<u>3.53</u>	0.66 ± 0.23	2.47
LDDMM (2008)	<u>0.72 ± 0.2</u>	4.53	<u>0.7 ± 0.21</u>	4.2	0.57 ± 0.25	5.0	0.4 ± 0.25	7.13
TS-LDDMM (ours)	0.83 ± 0.18	2.93	0.8 ± 0.18	2.07	0.7 ± 0.26	3.33	0.51 ± 0.27	5.67

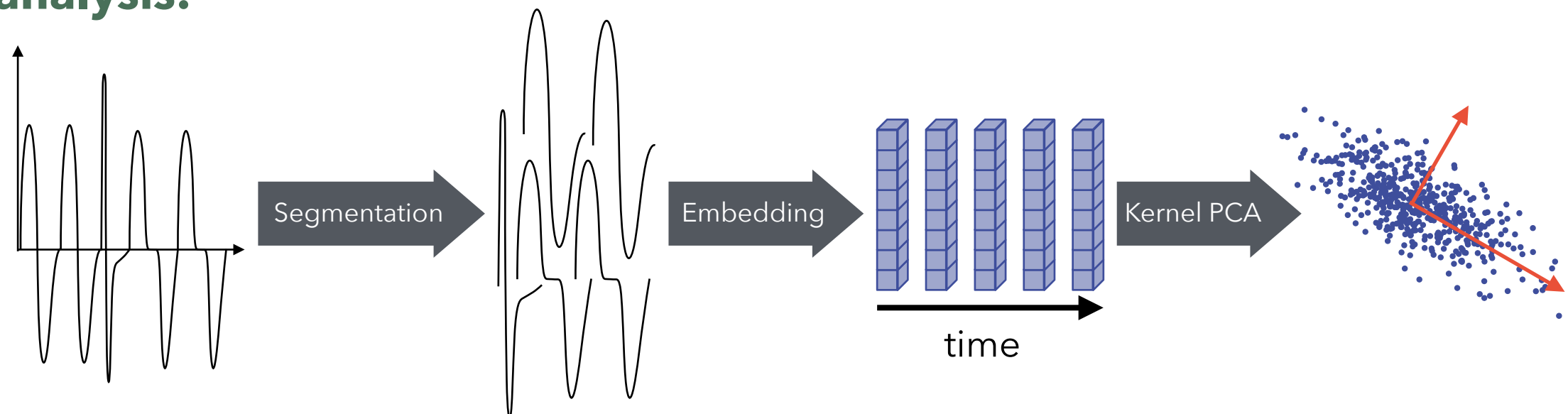
Mouse ventilation analysis

Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes (WT and ColQ).

Experimental protocol: For each genotype (WT, ColQ):

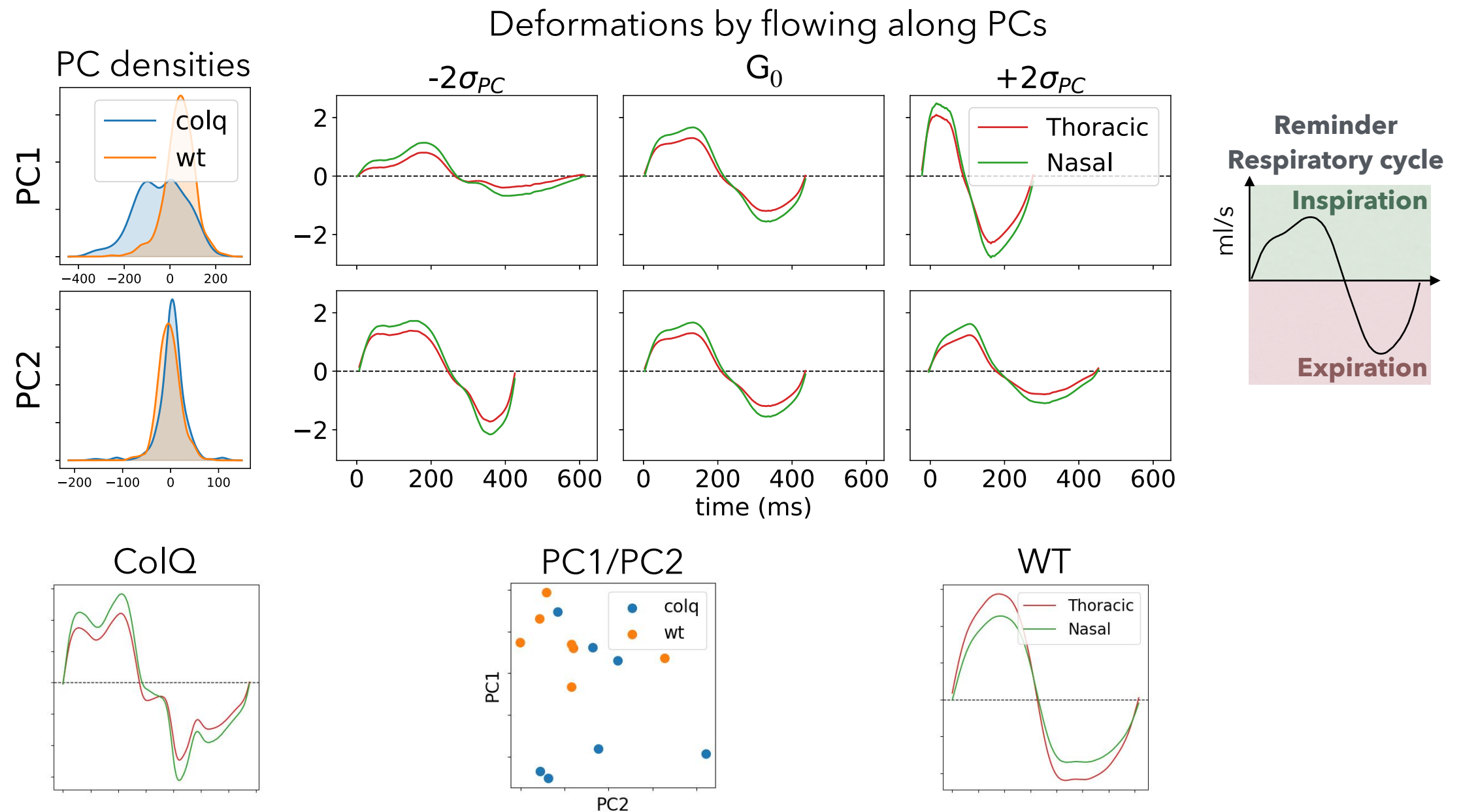


Data analysis:



Control period analysis

Kernel PCA applied on the initial velocity fields parametrized with $(\mathbf{G}_0, \alpha_j)_{j \in [N]}$.

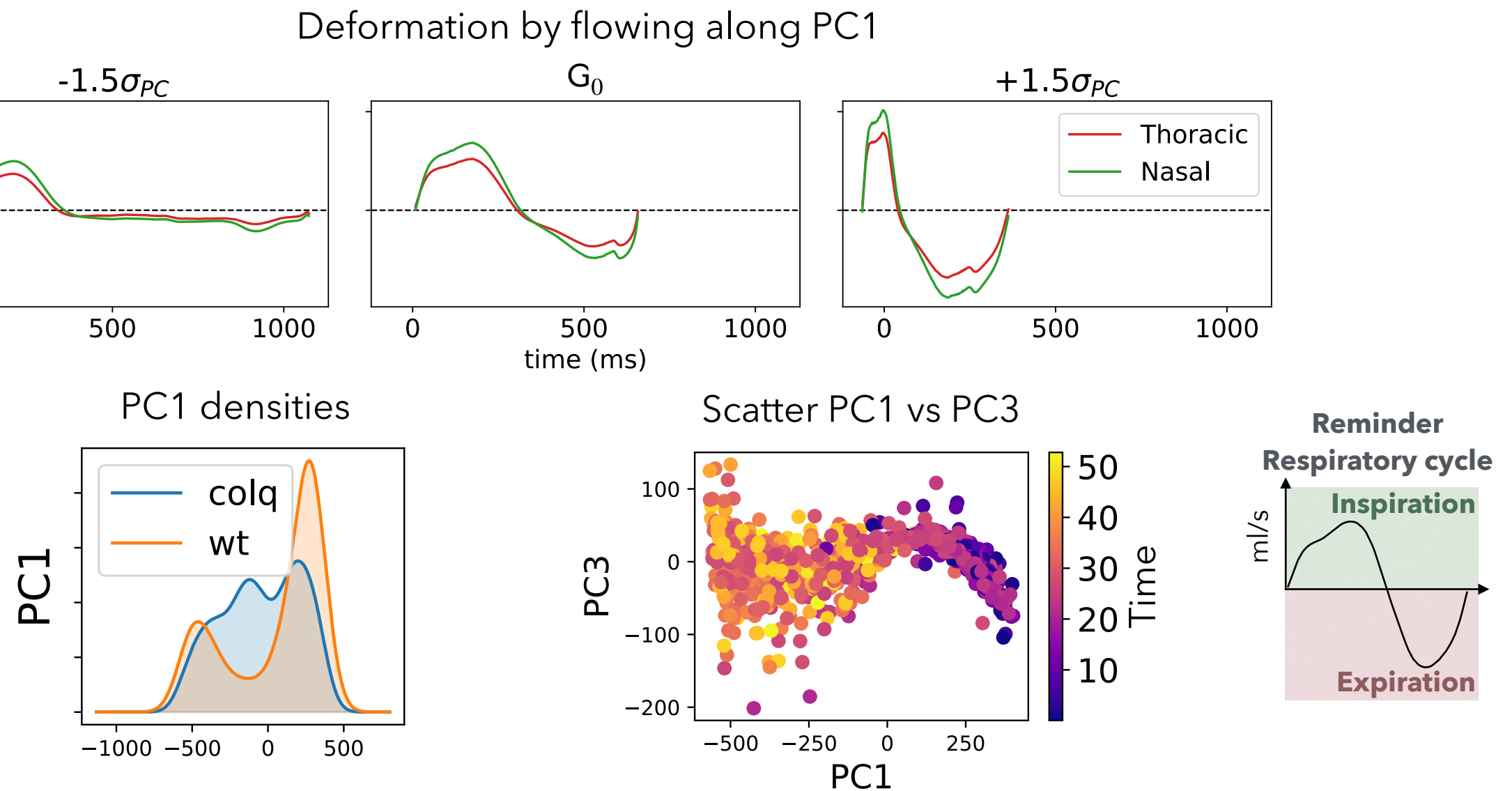


Observations:

- PC1 accounts for the cycle duration and PC2 for the ratio inspiration/expiration.
- PC1 negative tail for ColQ mice seems to highlight impairment of motor control.

Whole exercise analysis

Kernel PCA applied on the initial velocity fields parametrized with $(\mathbf{G}_0, \alpha_j)_{j \in \llbracket 1, N \rrbracket}$.

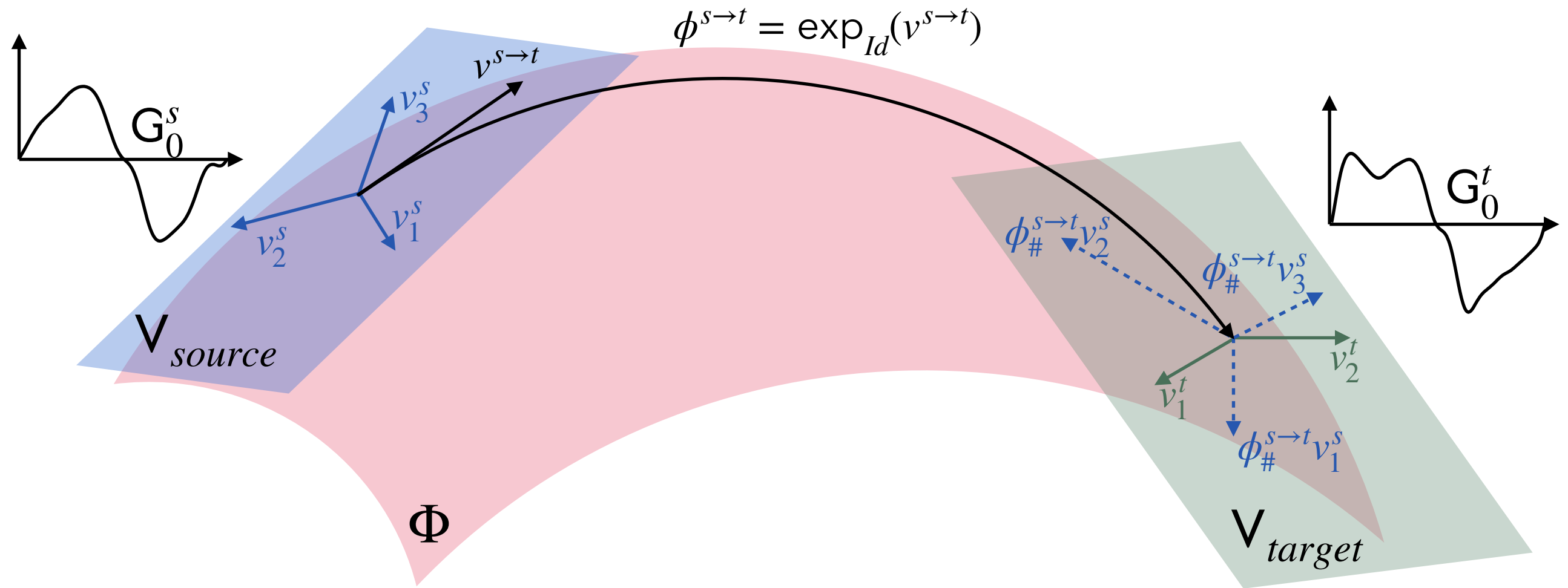


Observations:

- PC1 negative tail indicates a pause between inspiration and expiration.
- WT mice are more affected than ColQ ones after drug injection.
- Diversity of responses to the drug injection (scatter plot).

Going further

Domain Adaptation



Parallel transport: a tool to transport vectors from one tangent space to another. Exist on the space of diffeomorphisms Φ [BE05].

Suggestion:

Input: source and target reference graphs, a predictor learned on source vectors.

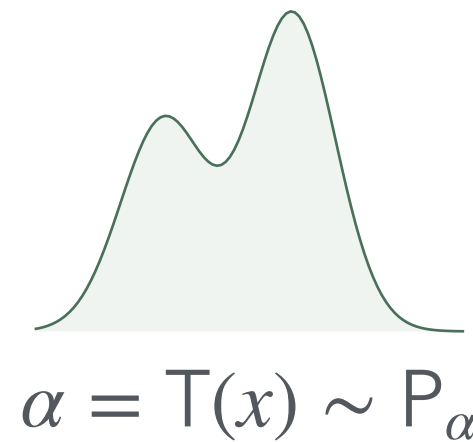
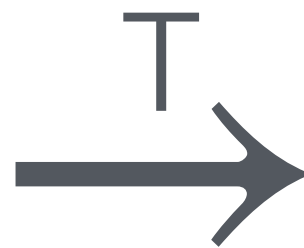
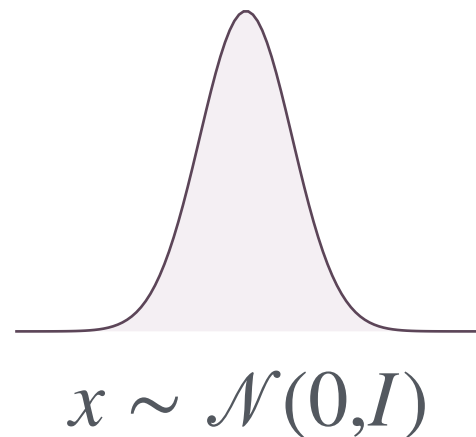
1. Perform the registration between the source and target reference graphs: $\phi^{s \rightarrow t}$
2. Parallel transport source vectors

Output: source predictor adapted to target domain

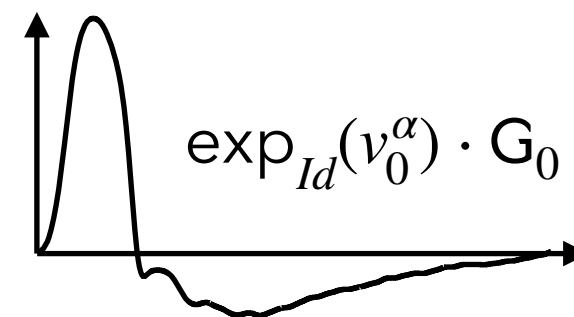
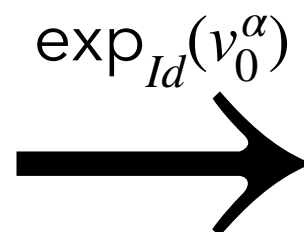
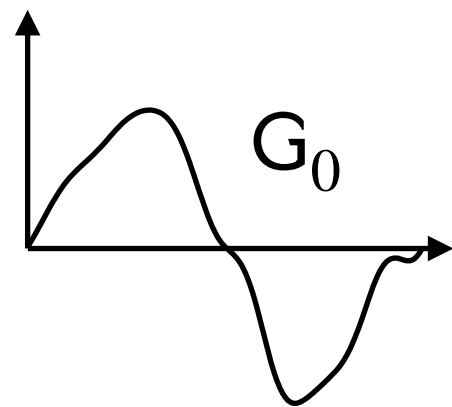
Generative Modeling

Recall: the embedding learning outputs a graph $G_0 \in \mathbb{R}^{m \times (d+1)}$, and momentums $(\alpha_k^0)_{k \in [N]} \in (\mathbb{R}^{m \times (d+1)})^N$.

Let:



Then:



Suggestion: Suppose that $\alpha \sim P_\alpha$.

1. Learn T such that $T_\# \mathcal{N}(0, I) = P_\alpha$.
2. Sample x from $\mathcal{N}(0, I)$.
3. Set $\alpha = T(x)$.
4. Perform a geodesic shooting from G_0 with initial velocity field v_0^α .

Thank you